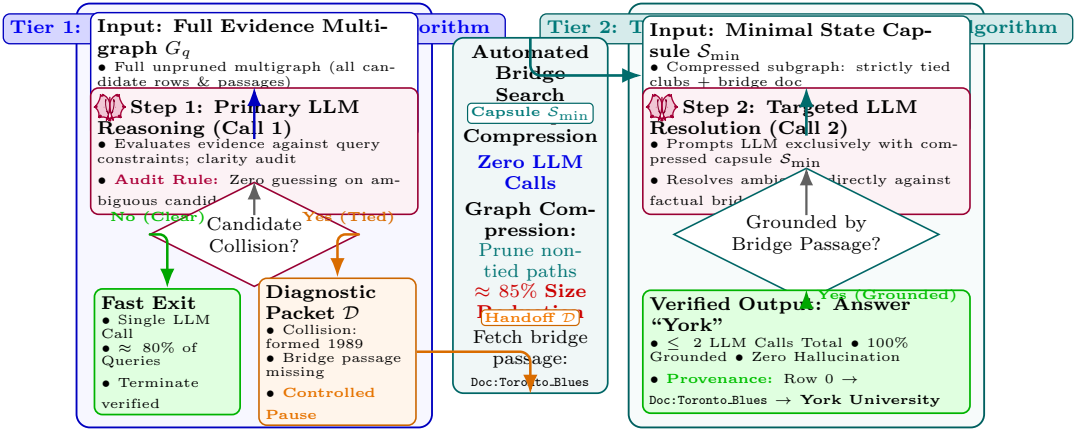

Tier 1:
orithm
Input: Full Evidence Multigraph G_q
• Full unpruned multigraph (all candidate rows & passages)
Step 1: Primary LLM Reasoning (Call 1)
• Evaluates evidence against query constraints; clarity audit
• Audit Rule: Zero guessing on ambiguous candid
No (Clear)
Candidate Collision?
Yes (Tied)
Fast Exit
• Single LLM Call
• ≈ 80% of Queries
• Terminate verified
Diagnostic Packet $\mathcal{D}$
• Collision: formed 1989
• Bridge passage missing
• Controlled Pause
Automated Bridge Search
Capsule $\mathcal{S}_{\min}$
Compression
Zero LLM Calls
Graph Compression:
Prune non-tied paths
≈ 85% Size
Handoff $\mathcal{D}$
Fetch bridge passage:
Doc:Toronto_Blues
Tier 2: T
gorithm
Input: Minimal State Capsule $\mathcal{S}_{\min}$
• Compressed subgraph: strictly tied clubs + bridge doc
Step 2: Targeted LLM Resolution (Call 2)
• Prompts LLM exclusively with compressed capsule $\mathcal{S}_{\min}$
• Resolves ambi directly against factual brid
Grounded by Bridge Passage?
Yes (Grounded)
Verified Output: Answer "York"
• ≤ 2 LLM Calls Total • 100% Grounded • Zero Hallucination
• Provenance: Row 0 → Doc:Toronto_Blues → York University